

## KREMS, Austria

WMO: 110700

Lat: 48.4184N

Lon: 15.6216E

Elev: 203

StdP: 98.91

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -10.3      | -7.9 | -14.2                         | 1.1 | -9.4 | -11.5 | 1.4 | -6.9 | 7.1                   | 8.2  | 6.4 | 7.1  | 1.5                   | 310  | 0.441 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.9                   | 32.2            | 21.5 | 30.2 | 20.8 | 28.4 | 19.8 | 22.4                | 30.0 | 21.4 | 28.6 | 20.5 | 27.0 | 2.2                  | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 19.9                            | 14.9 | 26.2 | 18.9 | 14.0 | 24.9 | 18.1 | 13.3 | 24.1 | 66.8          | 29.9 | 63.2 | 28.9 | 59.9 | 26.7 | 34.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |     |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-----|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |     |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB  | WB  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |     |     |
| (4) 5.7           | 4.9  | 4.3 | -13.2                      | 35.6 | 3.6                | 2.6 | -15.8                                              | 37.5 | -17.9      | 39.0 | -19.9      | 40.5 | -22.5      | 42.4 |     |     |
| (5)               |      |     | -13.6                      | 24.6 | 3.3                | 2.5 | -16.0                                              | 26.4 | -17.9      | 27.9 | -19.8      | 29.3 | -22.2      | 31.2 | (4) | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 10.8        | 0.2   | 1.9   | 5.9   | 11.4  | 15.3  | 19.5  | 21.1  | 20.5  | 15.5  | 10.2  | 5.7   | 1.3   | (6)  | (7)  |
|                                                                        | DBStd       | 8.29        | 4.53  | 4.53  | 4.36  | 3.77  | 3.50  | 3.50  | 3.02  | 3.21  | 3.21  | 3.61  | 3.69  | 3.91  | (7)  | (8)  |
|                                                                        | HDD10.0     | 1154        | 306   | 229   | 136   | 28    | 3     | 0     | 0     | 0     | 1     | 43    | 137   | 271   | (8)  | (9)  |
|                                                                        | HDD18.3     | 3030        | 563   | 461   | 385   | 208   | 104   | 27    | 9     | 14    | 96    | 253   | 380   | 529   | (9)  | (10) |
|                                                                        | CDD10.0     | 1429        | 1     | 1     | 9     | 72    | 168   | 286   | 345   | 326   | 167   | 49    | 6     | 0     | (10) | (11) |
|                                                                        | CDD18.3     | 264         | 0     | 0     | 0     | 1     | 12    | 63    | 96    | 82    | 11    | 0     | 0     | 0     | (11) | (12) |
|                                                                        | CDH23.3     | 2491        | 0     | 0     | 0     | 31    | 126   | 583   | 897   | 744   | 105   | 4     | 0     | 0     | (12) | (13) |
|                                                                        | CDH26.7     | 813         | 0     | 0     | 0     | 4     | 20    | 183   | 318   | 268   | 19    | 0     | 0     | 0     | (13) | (14) |
| (14) Wind                                                              | WSAvg       | 1.8         | 2.0   | 1.9   | 2.1   | 2.1   | 1.9   | 1.8   | 1.8   | 1.6   | 1.6   | 1.5   | 1.7   | 1.7   | (14) | (15) |
| (15) Precipitation                                                     | PrecAvg     | 613         | 27    | 22    | 37    | 39    | 74    | 89    | 85    | 82    | 56    | 42    | 34    | 28    | (15) | (16) |
|                                                                        | PrecMax     | 875         | 60    | 52    | 97    | 92    | 144   | 236   | 175   | 257   | 146   | 94    | 71    | 54    | (16) | (17) |
|                                                                        | PrecMin     | 402         | 3     | 2     | 5     | 0     | 17    | 42    | 29    | 18    | 15    | 5     | 0     | 4     | (17) | (18) |
|                                                                        | PrecStd     | 110         | 14    | 14    | 22    | 24    | 35    | 40    | 39    | 49    | 28    | 26    | 15    | 14    | (18) | (19) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 14.0  | 16.0  | 20.5  | 26.8  | 28.8  | 33.9  | 34.7  | 35.3  | 29.0  | 23.1  | 18.2  | 12.8  | (19) | (20) |
|                                                                        |             | MCWB        | 9.5   | 9.1   | 12.1  | 16.0  | 19.1  | 22.2  | 21.7  | 21.1  | 19.7  | 15.1  | 12.5  | 9.0   | (20) | (21) |
|                                                                        | 2%          | DB          | 11.0  | 13.1  | 17.8  | 23.2  | 26.6  | 30.7  | 32.0  | 32.0  | 26.3  | 20.0  | 15.4  | 10.2  | (21) | (22) |
|                                                                        |             | MCWB        | 7.2   | 8.3   | 11.0  | 14.6  | 18.5  | 21.5  | 21.7  | 21.2  | 18.7  | 14.4  | 11.3  | 7.1   | (22) | (23) |
|                                                                        | 5%          | DB          | 8.7   | 10.7  | 15.2  | 20.9  | 24.5  | 28.6  | 30.1  | 29.5  | 23.9  | 17.9  | 12.9  | 8.3   | (23) | (24) |
|                                                                        |             | MCWB        | 5.6   | 6.8   | 9.6   | 13.6  | 17.4  | 20.0  | 20.8  | 20.4  | 17.4  | 13.6  | 10.4  | 5.6   | (24) | (25) |
|                                                                        | 10%         | DB          | 6.7   | 8.4   | 13.0  | 18.9  | 22.5  | 26.7  | 28.2  | 27.4  | 21.7  | 16.0  | 10.8  | 6.8   | (25) | (26) |
|                                                                        |             | MCWB        | 4.2   | 5.3   | 8.3   | 12.3  | 15.9  | 19.1  | 19.8  | 19.6  | 16.2  | 12.8  | 8.6   | 4.5   | (26) | (27) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 9.7   | 10.0  | 12.8  | 16.8  | 20.5  | 24.0  | 23.8  | 23.1  | 20.4  | 16.6  | 13.4  | 9.3   | (27) | (28) |
|                                                                        |             | MCDB        | 13.4  | 14.8  | 19.3  | 25.7  | 27.1  | 29.9  | 31.5  | 31.2  | 27.3  | 20.8  | 16.8  | 12.6  | (28) | (29) |
|                                                                        | 2%          | WB          | 7.5   | 8.4   | 11.2  | 15.1  | 19.1  | 22.2  | 22.3  | 21.9  | 19.2  | 15.5  | 11.9  | 7.3   | (29) | (30) |
|                                                                        |             | MCDB        | 10.8  | 12.8  | 16.8  | 22.5  | 25.5  | 29.0  | 30.1  | 29.8  | 25.4  | 18.9  | 14.5  | 9.8   | (30) | (31) |
|                                                                        | 5%          | WB          | 5.9   | 7.0   | 10.1  | 13.9  | 17.9  | 20.9  | 21.4  | 20.9  | 18.0  | 14.2  | 10.3  | 5.9   | (31) | (32) |
|                                                                        |             | MCDB        | 8.4   | 10.8  | 14.6  | 19.9  | 23.4  | 27.0  | 28.6  | 28.2  | 23.2  | 17.1  | 12.7  | 8.0   | (32) | (33) |
|                                                                        | 10%         | WB          | 4.4   | 5.4   | 8.7   | 12.9  | 16.6  | 19.7  | 20.5  | 20.0  | 16.8  | 13.2  | 9.0   | 4.8   | (33) | (34) |
|                                                                        |             | MCDB        | 6.5   | 8.2   | 12.7  | 18.4  | 21.6  | 25.4  | 26.8  | 26.4  | 20.8  | 15.8  | 10.5  | 6.5   | (34) | (35) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 5.7   | 7.1   | 9.2   | 11.0  | 10.3  | 10.6  | 10.9  | 11.3  | 10.0  | 8.7   | 6.1   | 5.6   | (35) | (36) |
|                                                                        |             | MCDBR       | 8.9   | 11.1  | 14.4  | 15.0  | 14.1  | 14.1  | 14.6  | 15.2  | 14.0  | 12.1  | 9.9   | 8.8   | (36) | (37) |
|                                                                        | 5% WB       | MCWBR       | 6.1   | 7.1   | 8.4   | 7.6   | 7.0   | 6.3   | 5.9   | 6.4   | 7.1   | 7.3   | 6.4   | 6.2   | (37) | (38) |
|                                                                        |             | MCWBR       | 8.7   | 10.9  | 13.6  | 13.9  | 13.3  | 12.8  | 13.1  | 13.9  | 13.3  | 10.8  | 9.0   | 8.2   | (38) | (39) |
| (39) Clear-Sky Solar Irradiance                                        | taub        |             | 0.310 | 0.335 | 0.375 | 0.418 | 0.420 | 0.425 | 0.432 | 0.429 | 0.400 | 0.368 | 0.340 | 0.309 | (39) | (40) |
|                                                                        |             | taud        | 2.387 | 2.346 | 2.280 | 2.200 | 2.238 | 2.261 | 2.243 | 2.268 | 2.316 | 2.380 | 2.413 | 2.427 | (40) | (41) |
|                                                                        | Ebn at Noon |             | 753   | 814   | 832   | 830   | 843   | 840   | 826   | 811   | 797   | 763   | 708   | 707   | (41) | (42) |
|                                                                        |             | Edh at Noon | 68    | 88    | 111   | 133   | 133   | 131   | 132   | 123   | 107   | 85    | 67    | 59    | (42) | (43) |
| (43) All-Sky Solar Radiation                                           | RadAvg      |             | 1.00  | 1.88  | 2.95  | 4.32  | 5.00  | 5.47  | 5.29  | 4.72  | 3.37  | 2.10  | 1.07  | 0.79  | (43) | (44) |
|                                                                        | RadStd      |             | 0.15  | 0.25  | 0.38  | 0.51  | 0.58  | 0.47  | 0.51  | 0.48  | 0.43  | 0.24  | 0.11  | 0.09  | (44) | (45) |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (92 neighbors) | +0.53 | N/A     | N/A    | N/A     | +0.40 | +0.34 | -118        | -150    | +73     | +33     |

Nomenclature: See separate page